

GLASS



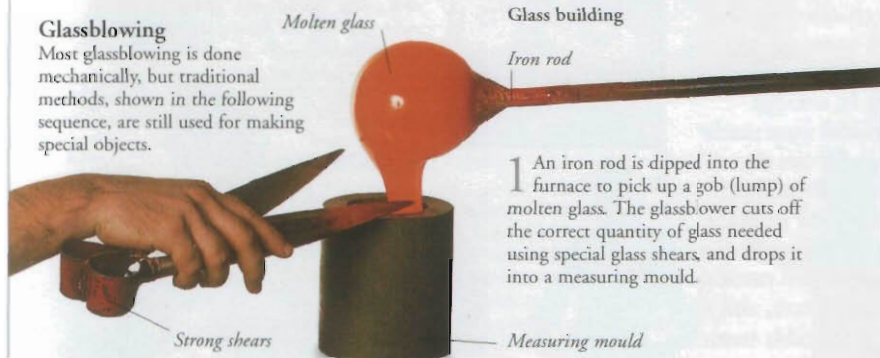
FEW MATERIALS have the same remarkable properties as glass. It is transparent, easy to shape and clean, does not rot, and resists attack by most chemicals. Glass is also cheap to produce because it is made from sand, one of the most common materials on Earth. When sand is heated with other materials, it turns into a liquid, which, when cooled, solidifies into glass. Although the glass looks crystalline, it still has the structure of a liquid, and is termed a "supercooled" liquid.

Working glass

Glass is easy to work, but only when it is in a molten state. The most common method of shaping glass is blowing by craftworkers or machines. Other methods include pressing molten glass into a mould, a traditional technique still used today, and casting it into a mould to make lenses.

Glassblowing

Most glassblowing is done mechanically, but traditional methods, shown in the following sequence, are still used for making special objects.



Glass building

Sheet glass

Sheet glass was originally made by drawing a ribbon of molten glass vertically upwards. However, this caused distortion. Today, it is made by floating molten glass on a bath of molten tin. This float glass is of even thickness throughout and shows no distortion.

Shaping mould into which the shaped parison is fitted.



Blowing glass

2 The glassblower picks up the molten glass from the measuring mould on a blowing-iron, then blows air through the iron, to shape the glass and form a parison.

3 With further blowing, the glass expands and takes its final shape inside the mould. At the same time, the rod is spun to stop the object showing signs of joints from the mould.



Layers of steam protect and cushion the glass.



The bottle shows no signs of the joint between the two halves of the mould.



Ancient glass

Decorative glass objects have been found in ancient Egyptian tombs dating back to 2500 BC. After the invention of the blowpipe in about 100 BC, glass was made across the ancient world, particularly in Rome.

Roman glassware, dating from 1st century AD

Types of glass

Three main ingredients are used to make glass: pure silica sand, soda ash, and lime. These are heated in a furnace to about 1,400 °C (2,500 °F), to produce soda-lime glass. This is the ordinary glass we use to make bottles and windows.

Different kinds of glass can be made by adding other ingredients.



Lead crystal



Spectacles



Magnifying glass



Heat-resistant glass jug

Lead glass

Also called crystal glass, lead glass contains lead oxide, which makes it easy to cut. The cut glass exhibits a diamond-like sparkle.

Optical glass

Optical lenses are made from pure glass. A variety of substances, such as lead and titanium, are added to give glass its optical properties.

Heat-resistant glass

Boron oxide is sometimes added in the glassmaking process to produce heat-resistant borosilicate glass.

Fibreglass

Glass may sometimes take the form of fibres, which are used for loft insulation, reinforcement for plastics, and fibre-optic cables.

Stained glass

Stained glass is normally used to create decorative windows, using pieces of coloured glass set in a lead framework. Medieval stained glass may seem richer to the eye because it is full of impurities.



Stained-glass window



Bottles ready to be recycled

Recycling glass

Glass is an easy material to recycle because it melts readily. It is recycled, not for the purpose of conservation, but to save energy because the original glassmaking process requires such high temperatures. In Europe alone, about four million tonnes of glass are recycled each year.

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